Contents

1 Open Science Reproducible 1
1.1 What can we learn from software engineering? 2

1 Open Science Reproducible

- Open science movement: @NosekPromotingOpenResearch2015; @MunafoManifestoReproducibleScience2017

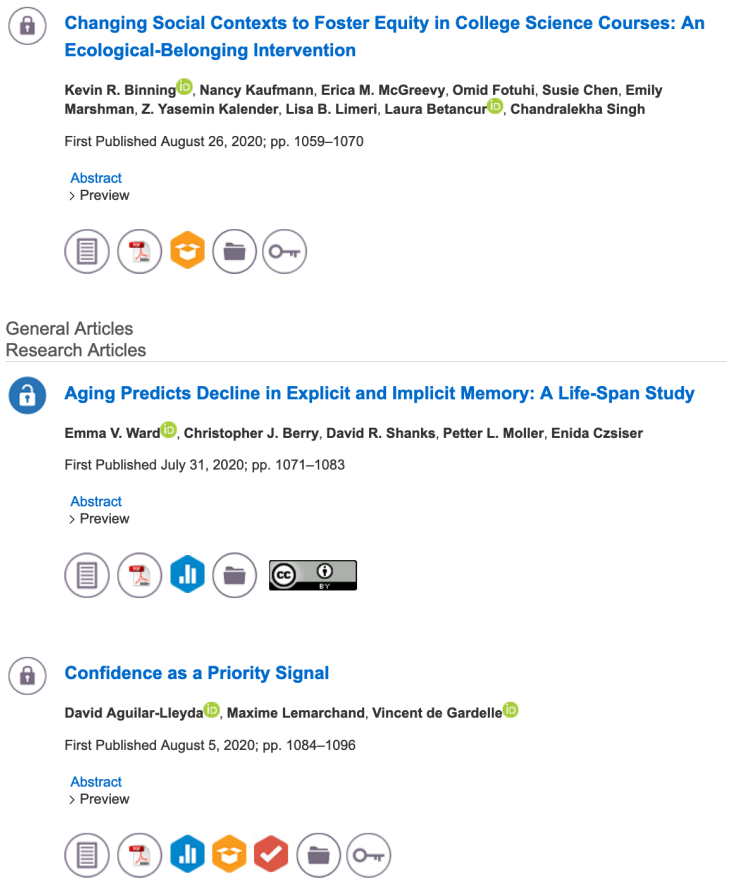
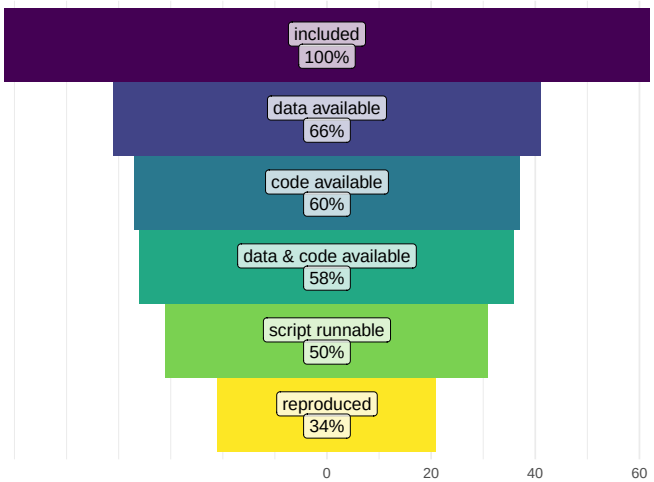


Figure 1: As part of the open science movement, some journals have added badges to journal articles indicating open data, open code, etc.

- @ObelsAnalysisOpenData2020
  - Reproducibility check of 62 pre-registered studies in a Center for Open Science database
  - "Of the 62 articles that met our inclusion criteria, 41 had data available, and 37 had analysis scripts available"
  - "We could run the scripts for 31 analyses, and we reproduced the main results for 21 articles."



\begin{figure} \caption{Infographic for @ObelsAnalysisOpenData2020. While 58% of preregistered studies provided both the data and analysis script, only 50% had runnable scripts and only 34% could actually be reproduced.} \end{figure}

- An example: https://osf.io/k5qmw/; @ColomboIntellectuallyHumblePrejudiced2020
- Better: https://osf.io/he5za/; @CollingRegisteredReplicationReport2020
  - Although:
  - Binder stalled out my first attempt
  - The manuscript file can't be knit
  - Running the analysis requires finding and running a series of scripts
  - Some of the scripts have warnings and errors, and it's not clear if those are problems

1.1 What can we learn from software engineering?

```
- version control
- good coding style
    - <https://style.tidyverse.org/>
    - prefactoring
    - <https://www.youtube.com/watch?v=7oyiPBjLAWY>
- code review [@PostolovskiYourCodeReview]
- documenting code (and data)
    - self-documenting code: <https://dev.to/woubuc/self-documenting-is-a-myth-and-how-
to-make-your-code-self-documenting-3h2n>
- the life-changing magic of tidying up (your files)
    - @WilsonBestPracticesScientific2014; @WilsonGoodEnoughPractices2017; @NobleQuickGuideOrganizing200
    - my project template
- later: automation, CI
```

- lab/exercise: spend 90 minutes on EDA of the Silberzahn et al. red card data, then code review